Fairfax Invitational - Repaired practice edition

Division C | Forensics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

chemistry-7. (1 points) Authored replacement for source item chemistry-7: Repair data: a sodium carbonate solution has pH 11.2; a calcium nitrate solution has pH 6.8. Which of these two solutions is basic, and what threshold did you use?

Answer: _____
_____
_____

mass-1. (1 points) Authored replacement for source item mass-1: The mass spectrum has its tallest peak at m/z 31 and a smaller molecular-ion peak at m/z 46. What is the m/z of the base peak?

Answer: _____
_____
_____

mass-2. (1 points) Authored replacement for source item mass-2: For a singly charged molecular ion with m/z 46 and a fragment base peak at m/z 31, what is the nominal molecular mass?

Answer: _____
_____
_____

repair-case. (4 points) Authored written substitute for unavailable source material: Why might a well-supported multilocus DNA comparison be more discriminating than a common fiber-class agreement?

Answer: _____
_____
_____